MILITARY SPECIFICATION

STEEL, CHROME-NICKEL-MOLYBDENUM, 8620, BARS, FORGING, AND TUBING, BEARING QUALITY, VACUUM DEGASSED

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a bearing quality, vacuum degassed, low alloy, chrome-nickel-molybdenum steel (8620) in the form of bars, forgings, forging stock, and tubing.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

STANDARDS
FEDERAL

FED-STD-48	- Tolerances for Steel Wrought Products, and for Centrifugally Cast Steel.
FED-STD-151	- Metal, Test Methods.
FED-STD-183	- Continuous Identification Marking of Iron and Steel-base Alloys.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 9510

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MILITARY

MIL-STD-163 - Steel Mill Products Preparation for Shipment and Storage.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN IRON AND STEEL INSTITUTE

Steel Products Manual
Alloy, Carbon and High Strength Low Alloy Steels:
Semifinished For Forging; Hot Rolled Bars, Cold Finished Bars
Hot Rolled Deformed and Plain Concrete Reinforcing Bars

(Application for copies should be addressed to American Iron and Steel Institute, 1000 16th Street, N.W., Washington, D.C. 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A255	- End-Quench Test for Hardenability of Steel, Method of.
ASTM A370	- Methods and Definitions for Mechanical Testing of Steel Products.
ASTM E45	- Determining the Inclusion Content of Steel, Practice for.
ASTM E112	- Estimating the Average Grain Size of Metals, Methods for.
ASTM E350	- Chemical Analysis of Carbon Steel, Low-Alloy Steel, Silicone Electrical Steel, Ingot Iron and Wrought Iron, Methods for.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.3).

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3.2 Material. The steel shall be vacuum degassed bearing quality.

3.2.1 Bars. Bars shall be furnished in accordance with applicable engineering drawings, in a machinable condition, hot finished or cold finished as specified (see 6.2) with a maximum Brinell hardness of 269 HBN (3000 kilogram load) or equivalent.

3.2.2 Forgings. Forgings shall be furnished in accordance with applicable engineering drawings and as specified herein (see 6.2).

3.2.3 Forging stock. Forging stock shall be furnished in accordance with applicable engineering drawings and as specified herein (see 6.2).

3.2.4 Tubing. Tubing shall be furnished in accordance with applicable engineering drawings, in a machinable condition, cold finished with a maximum hardness of Rockwell HRC 25 or equivalent, or hot finished with a maximum hardness of Rockwell HRB 99 or equivalent (see 6.2).

3.2.5 Chemical composition. Chemical composition shall be as specified in table I.

TABLE I. Chemical composition.

Elements	Percent 1/ 2/	Check analysis tolerances 3/			
		To 100 square inches	Over 100 to 200 square inches	Over 200 to 400 square inches	Over 400 to 800 square inches
Carbon	0.18 to 0.23	±0.01	±0.02	±0.03	±0.04
Manganese	0.70 to 1.00	±0.03	±0.04	±0.05	±0.0
Phosphorus	0.025	±0.005 4/	±0.01 4/	±0.01 4/	±0.01 4/
Sulfur	0.025	±0.005 4/	±0.01 4/	±0.01 4/	±0.01 4/
Silicon	0.20 to 0.35	±0.02	±0.02	±0.03	±0.04
Chromium	0.40 to 0.60	±0.03	±0.03	±0.04	±0.05
Nickel	0.40 to 0.70	±0.03	±0.03	±0.03	±0.03
Molybdenum	0.15 to 0.25	±0.02	±0.03	±0.03	±0.04
Copper	0.35	±0.02	±0.03	±0.03	±0.04

1/ All percentages are maximum unless shown as range or otherwise indicated.

2/ Applicable to sizes not exceeding 200 square inches in cross-sectional area, or 18 inches in width, or 10,000 pounds in weight per piece. For oversizes, chemical composition shall be specified by the acquisition activity.

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3/ Individual determinations may vary from the specified range to the extent shown in the various columns. Elements in any heat shall not vary both above and below the specified range.
4/ Over maximum only.

3.2.6 Grain size. Austenitic grain size shall be number five or finer. Occasional grains large as number three are permissible.

3.3 Hardenability. Hardenability of material shall meet the requirements of 3.3.1 unless the Government stipulates that hardenability in 3.3.2 shall apply.

3.3.1 End-quench hardenability. End-quench hardenability values shall be HRC 48 @ J1 maximum and HRC 32 @ J3 minimum for steel in all specified conditions using standard specimens processed in accordance with ASTM A255.

3.3.2 Specimen hardenability. Specimens with section 0.125 inch (in.) and 0.375 in. thick and not greater than 2 square inches (sq. in.) in area shall be cut from bar, forging, or tube after normalizing at 1700 degrees Fahrenheit (°F) ± 10°F. Specimen shall be ground and protected by suitable means or treated in an atmosphere to minimize scaling and eliminate either carburization or decarburization during heat treatment. Specimens shall be placed in a furnace at 1550 ± 25°F, allowed to heat to 1550 ± 25°F, held at temperature for 15 minutes, and quenched in commercial quenching oil. Each specimen shall have an average hardness of Rockwell, HRC 32 to HRC 47.

3.4 Macroetch. Specimens representative of cross section billets for forging or rerolling, shall be macroetched in hot hydrochloric acid (1:1) at 160 to 180°F for sufficient time to develop well defined macrostructure. Such specimens shall be free from pipe, excessive porosity, and segregation.

3.5 Microinclusions. Radial specimens, approximately 0.28 sq. in. in surface area, cut midway between center and surface of hardened samples, shall be polished on a face longitudinal to direction of rolling for microinclusion rating in accordance with the Jernkontoret Chart, plate 1, of ASTM E45. The worst field in any specimen shall not exceed limits of table II. For B and C type inclusions, rate field 2, when field contains a continuous inclusion less than 0.015 inches. Rate field 2-1/2, when field contains a continuous inclusion less than 0.25 inches. Rate field 3, when field contains a continuous inclusion less than 0.040 inches. A continuous inclusion is defined as space intervening between "in-line" inclusions and is less than 0.005 inches. Sample shall have no more than one number 3 field and no more than three number 2-1/2 fields, except in type "A", where 5 fields are permissible.

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TABLE II. Worst-field inclusion ratings.

	Inclusion ratings			
Type	A	B	C	D
Thick	2	2-1/2	1-1/2	1-1/2
Thin	3	3	2-1/2	2

3.6 Tolerances. Unless otherwise specified, tolerances shall meet the applicable requirements of the American Iron and Steel Institute, Steel Products Manual, for respective products.

3.6.1 Exact length tolerances. Bars or tubes of all sizes may be ordered to exact length or in lengths expressed as a multiple of a definite unit. Tolerances shall be in accordance with FED-STD-48 or as specified in the contract or purchase order (see 6.2).

3.6.2 Mill length tolerances. When exact or multiple lengths are not ordered or specified, bars or tubes shall be acceptable in mill lengths of 6 to 20 feet (ft.), but not more than 10 percent of any order as shipment shall be furnished in lengths shorter than 10 feet.

3.7 Identification marking. Individual bars, tubes, or bundles shall be identified in accordance with FED-STD-183. Marking shall include the number of specification, heat numbers of metal, and manufacturer's identification. Forgings and forging stock shall be identified as agreed upon by the Government and manufacturer.

3.8 Workmanship. Material shall be of a uniform quality and condition, clean and free from foreign materials and internal and external imperfections.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Materials. To determine conformance to 3.2, inspection, test, and material certification records shall be maintained by the contractor and shall be examined by the Government.

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4.3 Classification of inspections. Inspection requirements specified herein are classified as follows:

(a) First article (preproduction) inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).
(c) Acceptance test (see 4.5.3.1).
(d) Control test (see 4.5.3.2).

4.4 First article inspection. Two first article samples (see 3.1 and 6.3) shall be inspected at a location approved by the Government to determine conformance to requirements of this specification. Inspection shall consist of testing as specified in table III and examination as specified in table IV. All inspection shall be done by the contractor under Government surveillance. All tests and examinations shall be documented and made available to the Government upon request.

4.4.1 Failure. Failure of a first article sample to pass any examination or test specified may be cause for the Government to withdraw from surveillance of additional testing until faults revealed by the tests have been corrected.

TABLE III. Order of first article testing.

Description	Requirement	Test
Chemical composition	3.2.5	4.6.1
Grain size	3.2.6	4.6.2
Hardenability	3.3	4.6.3
End-quench hardenability	3.3.1	4.6.3
Specimen hardenability	3.3.2	4.6.3
Macroetch	3.4	4.6.4
Microinclusions	3.5	4.6.5
Tolerances	3.6	4.6.6
Exact lengths tolerances	3.6.1	4.6.6
Mill lengths tolerances	3.6.2	4.6.6

4.5 Quality conformance inspection.

4.5.1 Lot formation. Unless otherwise specified (see 6.2), a lot shall consist of all material identifiable to a single heat or melt. A lot shall consist of mill formed bars of one composition, shape, condition, and nominal dimension from a single heat or melt, heat treated as a batch or sequentially heat treated in a continuous furnace. The lot shall be submitted for inspection at one time.

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4.5.2 Quality conformance examination.

4.5.2.1 Sampling for quality conformance examination.

4.5.2.1.1 Sampling for product examination. For compliance with dimensions, identification, workmanship, and preparation for delivery requirements, samples of the product shall be drawn at random from the lot as indicated in table IV.

TABLE IV. Sampling for product examination.

Number of items in inspection lot	Number of items in sample (randomly selected)	Acceptable number (maximum number of sample items non-conforming to any test)
15 or less	All	0
16 to 180	15	0
181 to 300	35	0
301 to 500	50	1
501 and over	75	2

4.5.2.1.2 Sampling for chemical composition. A random sample from two or more finished bars, tubes, forging stock, or forgings shall be selected from each lot for chemical analysis in accordance with 4.6.1. Samples shall be taken parallel to the axis of the material selected at a point midway between the center and the surface. Material less than 1.25 in. thick shall be sampled throughout the entire cross section.

4.5.2.1.3 Sampling for grain size. One or more samples shall be selected to represent each heat of steel or inspection lot for determination of grain size in accordance with 4.6.2.

4.5.2.1.4 Sampling for hardenability, macroexamination, and microinclusions. A random sample shall be taken from each lot for each test to determine hardenability (see 4.6.3), macrostructure (see 4.6.4), and microinclusions (see 4.6.5). Forged or rolled samples shall be acceptable for the determination of hardenability.

4.5.2.2 Acceptable quality level. Each sample selected in accordance with 4.5.2.1 shall be examined for conformance to the acceptable quality levels (AQLs) in accordance with table IV on the basis of percent defective.

4.5.2.3 Classification of defects. For examination purposes, defects shall be classified as specified in table V.

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TABLE V. Classification of defects.

Category	Defect	Method of inspection
Minor:		
201	Dimensions not within tolerance (see 3.6, 3.6.1, and 3.6.2).	Gage
202	Incorrect or illegible marking (see 3.7).	Visual
203	Poor workmanship (see 3.8).	Visual

4.5.3 Quality conformance tests.

4.5.3.1 Acceptance test.

4.5.3.1.1 Sampling for acceptance testing. Samples for acceptance testing shall be selected in accordance with level S-3 of MIL-STD-105.

4.5.3.1.2 Acceptance testing. Each sample selected in accordance with 4.5.3.1.1 shall be subjected to the tests specified in table VI in the order listed, using an AQL in accordance with table IV on the basis of percent defective.

TABLE VI. Acceptance tests.

Description	Test
Chemical composition	4.6.1
Grain size	4.6.2
Macrostructure	4.6.4
Tolerances	4.6.6
Exact length tolerances	4.6.6
Mill length tolerances	4.6.6

4.5.3.1.3 Failure. Failure to pass acceptance examination or testing shall be cause for the Government to refuse to accept subsequent lots until it has been proven to the satisfaction of the Government that any deficiency has been corrected.

4.5.3.2 Control tests. The contractor shall conduct control tests to ensure the quality of the product. Such tests shall be subject to Government surveillance. All testing shall be documented and made available to the Government upon request.

4.5.3.2.1 Sampling for control testing. Control test samples shall be selected at the rate of 1 of each 100 units produced, except

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that not more than 4 nor less than 2 shall be tested in any 30-day period.

4.5.3.2.2 Control testing. Samples selected in accordance with 4.5.3.2.1 shall be examined for the defects specified in table IV, and subsequently subjected to the tests specified in table VII.

TABLE VII. Control tests.

Description	Test
Hardenability	4.6.3
End-quench hardenability	4.6.3
Specimen hardenability	4.6.3
Microinclusions	4.6.5

4.5.3.2.3 Failure. Failure of a control test sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots until it has been proven to the satisfaction of the Government that any deficiency revealed by examination or test has been corrected.

4.6 Methods of inspection.

4.6.1 Chemical composition test. To determine conformance to 3.2.5, sample bars selected in 4.5.2.1.2 shall be tested in accordance with method 111 or method 112 of FED-STD-151. ASTM E350 or equivalent may be used for composition determination.

4.6.2 Grain size test. To determine conformance to 3.2.6, grain size samples selected in accordance with 4.5.2.1.3 shall be determined in accordance with ASTM A112 using the McQuaid-Ehn Test for the austenite grain formation.

4.6.3 Hardenability test. To determine conformance to 3.3 through 3.3.2, samples selected in accordance with 4.5.2.1.4 shall be prepared and tested for hardenability in accordance with ASTM A255. The hardenability test in accordance with ASTM A255 shall not be required on a product which will not yield a suitable sample. However, the steel from which the product is made shall conform to 3.3.2.

4.6.4 Macrostructure test. To determine conformance to 3.4, transverse sections shall be made from the samples selected in 4.5.2.1.3. Specimens shall be cut a sufficient distance from the hot-cut surfaces and from the ends of the selected bars. The disk specimens shall be prepared and deep-acid etched using a hot 1:1

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hydrochloric acid solution at 160 to 180°F for a sufficient time to develop a deep well-defined macrostructure. Sample shall be examined for conformance to 3.4.

4.6.5 Microinclusion test. To determine conformance to 3.5, radial specimens approximately 0.28 sq. in. in surface area between center and surface of hardened samples, shall be polished on a face longitudinal to the direction of rolling for microinclusion rating in accordance with Jernkontoret Chart, Plate 1, of ASTM E45.

4.6.6 Dimensional measurements. To determine conformance to 3.6 through 3.6.2, samples selected in accordance with 4.5.2.1.1 shall be measured for cross-sectional area, diameter, width, thickness, and length.

5. PACKAGING

5.1 Preservation. Bars specified as cold drawn, cold rolled, turned, ground, or polished shall be coated with a suitable non-staining corrosion preventive oil, conforming to Type A preservative as detailed in MIL-STD-163 (see 6.2).

5.2 Packaging and packing. The bars shall be properly separated by size and condition when prepared for delivery. Packaging and packing shall be Level A or Level C as specified (see 6.2).

5.2.1 Level A. Bars shall be packaged and packed in accordance with MIL-STD-163 for either cold-finished or hot-rolled alloy steel bars, as applicable.

5.2.2 Level C. Bars shall be packaged and packed in accordance with the usual commercial practices. The exterior type shipping container shall be of such type that will insure safe transportation of product at the lowest rate to the point of delivery, and shall meet the requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking. In addition to any special markings required by the contract or order (see 6.2), markings shall be in accordance with MIL-STD-163.

6. NOTES

6.1 Intended use. Steel covered by this specification is intended for use primarily for critical carburized parts and components which are medium stressed and require moderately high core hardness.

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable engineering drawing required (see 3.2.1 through 3.2.4).
(c) If cold or hot finishing is required (see 3.2.1 through 3.2.4).
(d) If stress relieving is required (see 3.3.2).
(e) Chemical composition, if applicable, for oversizes.
(f) Exact length or length units, and length tolerances if mill stock is not acceptable (see 3.6, 3.6.1, and 3.6.2).
(g) Levels of preservation and packing required (see 5.1 and 5.2).
(h) Special markings (see 5.3).

6.3 First article. First article samples shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for examination, tests, and approval of the first article (see 3.1 and 4.4).

6.4 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.5 Supersession data. This specification supersedes General Dynamics Internal Specification, DAX 13396C, 15 December 1982.

Custodian:
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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-62470(AT)	Steel, Chrome-Nickel-Molybdenum, 8620, Bars, Forging and Tubing, Bearing Quality, Vacuum Degassed

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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